

Production RAG v1 — Measured Results

EXP-NULL through EXP-010 · 202 OpenAI + Anthropic documents · n = 20 questions / 22 evidence spans · evaluation-first

0.475 frozen production baseline BM25, control chunks	0.775 best measured configuration BM25 + transformer RRF @512	6 hypotheses tested by controlled intervention	4 falsified and kept as engineering evidence
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WHAT THIS PROJECT IS

A deliberately small, evaluation-first retrieval system. Every change is a controlled intervention against 20 human-verified questions whose evidence is anchored *above* the chunk layer — on (version_id, section_path, char_start, char_end) , never on chunk ids — so the ground truth survives every re-chunking. Negative results are preserved rather than deleted. At n = 20 one question is 5 percentage points, so no significance is claimed anywhere.

1. Getting a baseline that works

experiment	macro recall	full	spans@10	note
EXP-000 lexical — <i>as shipped</i>	0.000	0/20	—	websearch_to_tsquery ANDs every token
EXP-000 lexical — BM25 (the fix)	0.475	9/20	10/22	frozen production baseline
EXP-001 dense — LSA, corpus-fitted	0.300	5/20	—	not a pretrained model
EXP-002 hybrid interleave	0.450	8/20	—	
EXP-003 RRF rrf_k=60	0.500	9/20	—	no gain over BM25 alone

The first measurement found the shipped lexical search returning **nothing at all**: websearch_to_tsquery ANDs every token, so a 16-token question demanded all 16 tokens in one chunk. OR-ing the terms and ranking with BM25 — so a rare identifier outweighs a common word — took recall from **0.000 to 0.475**. That remains the production baseline.

2. Four hypotheses about the document representation

experiment	macro recall	full	spans@10	outcome
EXP-005 chunk size × BM25	0.475	9/20	10/22	3 questions rescued — falsified
EXP-006 structural enrichment × BM25	0.475	9/20	10/22	Δ+0.000 — falsified
EXP-007 static pretrained dense (FastText)	0.425	8/20	9/22	weak instrument; below BM25
EXP-007C BM25 + FastText RRF	0.600	11/20	13/22	unexpected win — retrievers are complementary
EXP-008 chunk size × dense	0.400	8/20	—	interaction -0.050 — not supported

Chunk size was tested against BM25 (EXP-005) and against dense retrieval (EXP-008); structural enrichment was isolated in a 2×2 (EXP-006). All three failed. EXP-006 found the mechanism: prefixing every chunk with Provider: Anthropic inflated that term's document frequency from 3,289 to 12,028, which is exactly what BM25 is designed to discount.

EXP-007's own hypothesis failed too — static FastText scored below BM25 — but produced the project's first real gain by accident: **fusing the two retrievers** reached 0.600, because they fail on different questions.

3. A real transformer, and what it actually showed

experiment	macro recall	full	spans@10	outcome
EXP-009 transformer dense @256 (reference primary)	0.500	10/20	11/22	missed its preregistered bar
EXP-009 BM25 + transformer RRF @256	0.625	12/20	14/22	
EXP-009 transformer dense @512 (preregistered sensitivity)	0.575	11/20	13/22	cleared the bar
EXP-009 BM25 + transformer RRF @512	0.775	15/20	17/22	best measured; +6 vs BM25, zero regressions
EXP-010 transformer dense, encoder-aligned	0.575	11/20	13/22	$\Delta 0.000$ vs control chunks — falsified
EXP-010 BM25 + aligned transformer RRF	0.775	15/20	17/22	$\Delta 0.000$ vs EXP-009 best

THE ONE CONFIGURATION THAT CLEARLY BEAT THE BASELINE

BM25 + transformer @512 RRF, both on control chunks — 0.775 / 15-of-20, MRR 0.449. Net +6 cases over BM25 and +4 over the FastText fusion, with zero regressions in either comparison — the only zero-regression result in the project.

4. EXP-010: the mechanism story was wrong

EXP-009 concluded that *encoder visibility* drove the gain, since retrieval improved as the context window widened. EXP-010 tested that causally: hold the window at 512 and rebuild the chunks so every unit fits.

	control	encoder-aligned
chunks over the 512-token window	3,300 (23.22%)	0 (0%)
corpus token coverage	0.7610	1.0000
transformer dense recall	0.575	0.575
fused recall	0.775	0.775

TRUNCATION ELIMINATED ENTIRELY — RETRIEVAL DID NOT MOVE AT ALL

$\Delta 0.000$ on both comparisons, zero rescued, zero regressed. The measurement that explains it: **only 1 of 22 answers was actually outside the visible window**. Chunk truncation is not answer invisibility — 23% of *chunks* were cut, but answers sit near the top of the sections carrying them, so 21 of 22 were already fully visible. There was almost nothing left to fix, and **AN-007, the single case the hypothesis predicted, got worse when made visible** (rank 16 → 28).

5. Where the recall actually is

Three chunking interventions have now returned **0**, **-0.050** and **0.000**. The document representation is not where the remaining recall lives:

- **21 of 22 answers are already visible** to the encoder, and 9 still sit outside the top 10. The encoder reads the right text and scores it below other text — that is a **ranking** problem, not a visibility one.
- **The query side has never been touched.** Every experiment so far changed the document representation; the query has been a raw user-question string since EXP-000.
- **A reranker is still not justified** — a perfect reranker over a 100-candidate pool tops out at 0.909 against 0.775 already delivered, roughly three spans of headroom.
- **AN-003 needs a failure report, not another chunker.** It has defeated BM25, FastText, the transformer at two windows, and encoder-aligned chunking — and its answer was visible every time.

6. Standing caveats

- $n = 20$ questions. One case is 5 percentage points; no significance is claimed and nothing here generalises beyond this corpus.
- **EXP-NUL (closed-book control) is BLOCKED** — no authorized generation credential is available, so we cannot yet measure how many of these questions a model answers with no retrieval at all. Until that runs, the retrieval gains are unanchored against a no-retrieval floor.
- The transformer bundle could not be checksummed against its upstream publisher (Hugging Face is blocked by egress policy); structural and behavioural checks were recorded instead.
- The frozen production baseline has **not** been changed. EXP-009's 0.775 rests on a single run and is a candidate, not a promotion.

Generated from experiments/**/*_json by scripts/build_project_summary_pdf.py. EXP-010 git commit 34dacc4d57e9, config hash dd050bde901cb3a6. Encoder sentence-transformers/all-MiniLM-L6-v2 @ 512 tokens, exact cosine, no ANN index. Retrieval carries no reranker, query rewriting, query expansion, stemming or enrichment. Raw provider documentation is not redistributed.